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Research Article

Social Sciences

Forest Owners' Intention to Safeguard Forest Biodiversity: An Application of the Theory of Planned Behavior

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Abstract

Private nonindustrial forest owners' intention to safeguard biodiversity in their own forest was studied by applying the theory of planned behavior. The data were collected in a nationwide mail survey sent to 3,000 Finnish forest owners (response rate 35%). The impact of attitudes, subjective norm, and perceived behavioral control on the intention to safeguard biodiversity was empirically tested by estimating structural equation models (SEM). The empirical estimations supported the theoretical model. The influence of perceived behavioral control was inhibiting on the intention and slightly stronger than the explanatory power of attitude or the subjective norm. External factors had an impact on the explanatory pattern of the intention. The effect of subjective norm increased and the role of attitude decreased with age. Academic education increased the importance of attitudes and decreased the role of normative pressures. In the development of voluntary measures aimed at safeguarding biodiversity, special attention should be paid to the reduction of the obstacles for the implementation perceived by forest owners. Forest owners often perceive important to retain decision-making power for the next generation. Forest owners should be provided with information about the different measures for safeguarding biodiversity and the contract terms.

Study Implications: Forest owners consider active timber production, compensation level, willingness to retain decision-making power for the next generation, and small forest ownership as factors that inhibit their intention to safeguard biodiversity in their own forests. Forest owners' attitude toward safeguarding forest biodiversity and the normative pressures the reference groups create have a slightly smaller effect on intention. It is important to take the views of forest owners into account when developing voluntary measures for safeguarding biodiversity. It is essential to distribute information to forest owners about the available measures, their requirements, and compensation.

Keywords: attitude, conservation, METSO program, private forest owners, structural equation model

Biodiversity has become one of the key factors in the management and use of forests. This trend has been especially enhanced by the EU Biodiversity Strategy for 2030 (European Commission 2021) and the European Green Deal (European Commission 2019). A focal question in halting the loss of biodiversity in private forests is how to encourage landowners to implement measures for safeguarding biodiversity. These measures, which aim to protect, enhance, or restore the forest's ecological value, include the establishment of permanent or fixed-term protection areas where forestry is not allowed, exclusion of a forest site or sites from timber harvesting by owner's decision, the use of forest management methods that take biodiversity specifically into account (e.g., leaving more retention trees in forest felling), and active nature management to restore and preserve forest biodiversity by the restoration of habitats or improving the living conditions of endangered species, for example. Private forest owners play an important role in safeguarding forest biodiversity because they own approximately half of European forests (Hirsch and Schmithüsen 2010). In Finland, about 60% of productive forest land is owned by private owners (Natural Resources Institute Finland 2019), and forest conservation in private forests is implemented mostly through the voluntary measures of the Forest Biodiversity Program for Southern Finland (METSO) (Government of Finland 2014). The main goal of the METSO program is to halt the decline of forest habitats and forest species and to stabilize the favorable development of biodiversity by 2025. The Ministry of the Environment and the Ministry of Agriculture and Forestry coordinate and finance the program. In the METSO program, landowners can voluntarily offer sites for temporary or permanent protection or enhance the biodiversity of their forests by participating in a nature management project. ELY Centers (the regional Centers for Economic Development, Transport and the Environment) implement the program with methods based on the Nature Conservation Act (2023) by establishing private nature reserves (ownership is retained, forestry is not allowed, compensation is mainly based on the timber value) by purchasing areas for the state for conservation purposes (the price is mainly based on the value of the timber and value of the land) and by establishing temporary nature reserves for 20 years (forestry is not allowed, the compensation is mainly based

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on the value of the economic loss incurred compared with management in terms of commercial forestry) (Government of Finland 2014). The Finnish Forest Center implements the METSO program with 10 year environmental forestry subsidy agreements in accordance with the Temporary Act on the Financing of Sustainable Forestry (2015). The subsidy agreement can be renewed if the conditions are still met after the 10 year contract period, and funding is available in the program. Compensation is based on the calculated value of the forest stand. Forestry is not allowed in the agreement area. In nature management projects, valuable natural features are restored and preserved by restoring mires or conducting controlled burning, for example. The Finnish Forest Center plans these projects. Participation in nature management projects does not create costs for the owner Government of Finland (2014). After a forest owner has offered a site to METSO program, the advisors of the Finnish Forest Center or the ELY Center inspect whether the offered site meets the program's selection criteria (Syrjänen et al. 2016). The selection criteria covers ten habitats important for forest biodiversity. Certain structural features, such as the abundance of decaying wood, endangered species, burnt or charred wood, and mature broad-leaved trees, increase the site's ecological value. The application phase does not bind the forest owner to entering a contract. The advisor and the forest owner negotiate the implementation measure and the delineation of the site, and a contract is signed if they agree (Government of Finland 2014; Syrjänen et al. 2016).

Knowing the attitudes and views of forest owners is important when developing voluntary measures for safeguarding biodiversity in private forests. Several studies have examined landowners' biodiversity values and attitudes. Positive environmental attitudes are related to participating in conservation programs (Comerford 2014; Mitani and Lindhjem 2015). Nordlund and Westin (2011) found that forest values had an effect on forest owners' forest management attitudes; ecological forest values had a negative effect on the economic attitude and a positive effect on both environmental and human-centered forest management attitudes.

Various actors can be sources of normative pressures for the owner in decision-making related to forests. The role of forestry advisors has been found important for forest owners' awareness of voluntary conservation (Korhonen et al. 2013). Vainio et al. (2018) found that forest owners sought biodiversity information more often from those stakeholders they felt to be trustworthy, and whose goals with which they felt able to identify. According to Fielding et al. (2005), stronger perceived normative support for riparian zone management was related to landowners' stronger intention to implement that practice.

Several factors can help or hinder forest owners in implementing measures for enhancing biodiversity. The contracting process's simplicity can support landowner participation in conservation programs (Whitten et al. 2013). Korhonen et al. (2013) have identified the importance of the ease of enrollment in the conservation program. According to Knoot et al. (2015), the program requirements play an important role in forest owners' willingness to produce forest ecosystem services, and increasing requirements decrease landowners' willingness to participate. The conditions of conservation measures affect forest owners' willingness to implement them (Koskela and Karppinen 2021), and compensation claims increase if the contract terms are undesirable from the owner's perspective (Horne 2006).

Previous experience has been found to be related to more frequent participation in other environmental programs or practices (Comerford 2014; Fielding et al. 2005). According to Primmer et al. (2014), previous experience in conservation contracting is related to forest owners' perceptions of the conservation's positive ecological impacts. The future intention to enter a contract was found to be associated with expectations of increased local and social welfare (Primmer et al. 2014).

Certain characteristics of landowners have been found to increase the probability of being involved in conservation or giving greater value to the ecological aspects of forest management. These include female gender (Nordlund and Westin 2011; Umaerus et al. 2019), high level of education (Comerford 2014; Drescher et al. 2017; Polome 2016), young age (Ulizcka et al. 2004), and urban residence (Häyrinen et al. 2015) or residence outside the forest property (Danley et al. 2021; Ingemarson et al. 2006). According to Ulizcka et al. (2004), self-estimated knowledge of conservation and forest animal and plant species is related to a positive attitude toward conservation. Ma et al. (2012) found that owners of larger forest properties were more likely to participate in forest conservation programs in the United States.

This study aims to describe the role of forest owner attitudes, normative pressures, and perceived controlling factors concerning decisions about conservation matters to explain forest owners' intention to safeguard biodiversity in their own forests during next 5 years. It applies the theory of planned behavior (TPB) (Ajzen 1985, 1991). The study also aims to clarify whether the background characteristics of the forest owner and forest property influence the explanatory pattern of attitude, normative pressures, and perceived controlling factors on the intention. The TPB comprehensively takes into account beliefs, attitudes, normative pressures, and constraining factors. Applying the TPB to studying forest owners' intention to protect biodiversity produces important information that can be used in the development of voluntary measures for safeguarding biodiversity.

Theoretical Background

The TPB was preceded by the theory of reasoned action (TRA) (Ajzen and Fishbein 1980; Fishbein and Ajzen 1975). In the TRA, intention is a function of one's attitude toward behavior and subjective norms. The TRA was later extended to include a new explanatory factor, perceived behavioral control, and was named the theory of planned behavior (Ajzen 1985, 1991; Ajzen and Driver 1992; Ajzen and Madden 1986) (figure 1). The TPB is based on the assumption that individuals make logical decisions to participate in certain behaviors. The theory suggests that intention is the causal antecedent of actual behavior and indicates a person's willingness and preparedness to perform a given behavior. It assumes that one's attitude, response to the desires of other people, and perceived control over a particular behavior form the intention to execute the behavior or not (Ajzen 1985, 1991; Ajzen and Driver 1992; Ajzen and Madden 1986). According to the TPB, the attitude toward a behavior is a product of behavioral beliefs about the effects of performing the behavior in question and the evaluation of the importance of the outcomes for the individual. The subjective norm is

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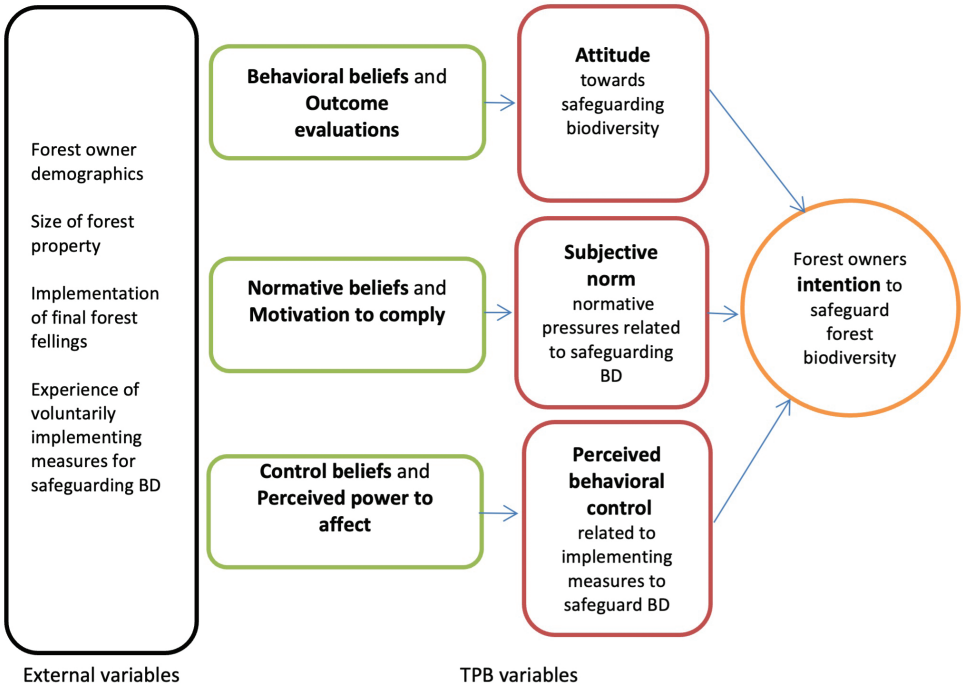


Figure 1 Factors affecting forest owners' intention to safeguard biodiversity in their forests. Modified from the TPB (Ajzen 1985, Ajzen 1991). TPB terms are presented in bold.

formed as a product of normative beliefs and the motivation to comply. Normative beliefs mean a person's perception about the judgment of relevant referents (social groups such as family, friends, advisors, colleagues) concerning whether the person should or should not conduct the behavior. The motivation to comply refers to a person's desire to behave in accordance with the referents' expectations. Perceived behavioral control is constructed as a product of control beliefs, which are a person's beliefs about the factors that facilitate or inhibit the performance of the behavior, and the perceived power to affect, which means the evaluation of how strongly the person perceives the controlling factors to affect the ease or difficulty of performing the behavior. It reflects the degree to which a person believes the decision to be under their own control (Ajzen 1985, 1991; Ajzen and Driver 1992; Ajzen and Madden 1986). Figure 1 shows this study's theoretical framework. The study examines the explanatory power of attitude, subjective norm, and perceived behavioral control regarding the forest owner's intention to safeguard their forests' biodiversity in the next 5 years. The effects of external factors on the explanatory pattern are studied by using subsamples of the data.

The TPB (Ajzen 1985; Ajzen 1991) has been applied to many research areas in the past and has also been found to be applicable to studying landowners' intentions concerning environmental issues (e.g., Deng et al., 2016; Fielding et al., 2005; Mastrangelo et al., 2014). Thompson and Hansen (2013) suggest that perceived behavioral control has the strongest relationship with forest owners' behavioral intentions to participate in carbon sequestration and trading. According to Karppinen (2005), forest owners' choice of reforestation method was mostly affected by attitude and past behavior; and Karppinen and Berghäll (2015) showed that the subjective norm was the strongest explanatory factor for forest owners' intention to conduct timber stand improvement. Among landowners in New England, normative pressures had the greatest influence on landowner intention to harvest trees threatened by invasive insects (Holt et al. 2021). For example, Sheeran's (2002) meta-analysis of ten previous meta-analyses suggested correlations between intentions and behaviors. In the study literature, intentions have been used to predict a wide range of behaviors; for example, academic activities and achievement (Manstead and van Eekelen 1998), smoking (Norman et al. 1999), and cancer screening attendance behaviors (Sheeran and Orbell 2000).

Materials and Methods

Survey Data

The study is based on a nationwide mail survey sent to 3,000 Finnish nonindustrial private forest owners (NIPF) in the fall of 2015. The Finnish Forest Center's (a state-funded organization that promotes forestry, advises landowners, and enforces forest legislation) forest owner register was used to select a random sample of forest owners owning at least 5 ha of forest land, including less productive forests. The sample included individuals owning forest land, people owning forest land with their spouse, private partnerships, and forest properties owned jointly by heirs. After two reminders, 1,035 satisfactorily completed questionnaires were received. The respondents were also given an opportunity to reply online, and 14% of the total responses were given through the online survey. The response rate of the survey was 35%. The English translations of the questions used in this study are presented in Supplement 1.

The average age of the respondents was 61 years, and 63% were male. The average forest property size was 43 ha. The occupational status of half the respondents (50%) was pensioner, one-third (32%) were wage earners, 9% were farmers or forestry entrepreneurs, 6% were other entrepreneurs,

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and 3% had another occupational status, such as student or unemployed. Half the respondents (52%) lived in a rural area, 17% in population centers, 17% in towns (>20,000 inhabitants), and 14% in cities (>100,000 inhabitants). Forty percent of the respondents lived permanently on the forest property, 21% elsewhere in the same municipality as the property, and 39% outside the municipality where the property was located. Two-thirds (66%) of the respondents owned the forest property independently or with their spouse, 22% were representatives of private partnerships, and 13% were representatives of an estate owned jointly by heirs. The characteristics of the respondents were rather similar to those in the recent Finnish Forest Owner Survey 2020 (Karppinen et al. 2020).

Nonresponse was studied through telephone interviews, with forty-six randomly chosen nonrespondents. The background characteristics of the nonrespondents were rather similar to those of the respondents. However, the share of farmers or forestry entrepreneurs was higher among the nonrespondents (24%) than respondents (9%). The nonrespondents supported increasing the level of safeguarding biodiversity in private forests less frequently (7%) than the respondents (19%), and the share of "cannot say" replies to that question was lower among the respondents (9%) than among the nonrespondents (27%). The percentage share of women was lower among the respondents (37%) than nonrespondents (49%). Among the nonrespondents, the average size of forest property was 51 ha, and forest owner mean age was 57 years. A more detailed description of the characteristics of the respondents and nonrespondents can be found in Juutinen et al. (2020) and Koskela and Karppinen (2021), which use the same dataset but mostly different variables than the current study.

The missing values of the variables used for structural equation modeling (SEM) were imputed using multiple imputation (Berglund and Heeringa 2014). The imputation percentage was 1%, and the effects of the imputation on the results were found to be insignificant. After the imputation process, there were 953 usable observations.

Variables

Intention

Respondents' intention to safeguard biodiversity in their own forest during the next 5 years (e.g., by establishing a permanent or fixed-term protection area, implementing nature management practices, or taking biodiversity specifically into account in forest management practices) was examined using a single 7-point Likert scale (very unlikely–very likely). The respondents who replied with 5 to 7 (likely to very likely) were named the forest owners with a strong intention, and the respondents who replied with 3 to 1 (unlikely to very unlikely) the forest owners with a weak intention to safeguard biodiversity. The respondents who replied with 4 were not included to this grouping.

Behavioral Beliefs and Outcome Evaluations

The strength of behavioral beliefs about the effects of the safeguarding biodiversity in forests and outcome evaluations of the importance of these aspects for the forest owners themselves were both measured with ten statements. Behavioral beliefs (Table 1) were measured using a 7-point Likert scale (very unlikely–very likely) and outcome evaluations (Table 1) using a 7-point Likert scale (completely unimportant–very important).

Table 1. List of the statements concerning behavioral beliefs about the effects of the safeguarding of biodiversity and the outcome evaluations of the importance of these aspects for the forest owner.

Behavioral beliefs	Outcome evaluations
Causes loss of income	Avoidance of loss of income
Increases opportunities for recreational use of forests	Recreational use of forests
Preserves nature for future generations	Preservation of nature for future generations
Preserves beautiful forest scenery	Beautiful forest scenery
Creates financial benefits for the forest owner	Financial benefit of conservation
Preserves endangered species	Preservation of endangered species
Produces positive impact on climate and waterbodies	Positive climate and waterbody impacts
Reduces local employment	Impacts on local employment
Reduces forest owner's property rights and power of decision	Retaining property rights and power of decision
Increases vitality of game species populations	Game species populations

Normative Beliefs and Motivation to Comply

Normative beliefs were examined by listing ten referent groups in forest-related matters (family members, local Forest Management Association advisors, other forest owners known by the respondent, forestry service entrepreneurs known by the respondent, e.g., a logger, neighbors, relatives and friends, local timber buyers, environmental advisors of the regional Center for Economic Development, Transport and the Environment (ELY), Finnish Forest Center advisors, representatives of nature conservation organizations). The respondent assessed the opinion of each referent group: Does the referent group expect the respondent to safeguard biodiversity in her/his own forests using a 7-point Likert scale ranging from "They think that I should not safeguard biodiversity" to "They think that I should safeguard biodiversity." The respondents also indicated separately for each of these referent groups their own motivation to comply with the referents' expectations using a 7-point Likert scale (not at all–completely).

Control Beliefs and Perceived Power to Affect

Control beliefs were studied with ten statements considering different factors that might facilitate or inhibit the safeguarding of forest biodiversity (Table 2). The strength of the control beliefs was measured using a 7-point Likert scale (completely disagree–completely agree). Perceived power to affect was assessed by asking the respondents how much each of these aspects affected their decision-making concerning safeguarding biodiversity in their own forests (a 7-point Likert scale, ranging from not at all–very much) (Table 2).

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Table 2. List of the statements concerning control beliefs about the factors that might facilitate or inhibit the safeguarding of forest biodiversity and perceived power to affect: How much these aspects affected forest owners' decision making related to safeguarding biodiversity.

Control beliefs	Perceived power to affect
Compensation for conservation is sufficient	Amount of compensation
Practical assistance is easily available for biodiversity matters	Availability of practical assistance
Suitable measures are available to safeguard biodiversity	Possibility of choosing suitable measure for safeguarding biodiversity
There are ecologically valuable sites in my forests	No ecologically valuable sites in my forests
Environmental authorities are reliable negotiation partners	Distrust in environmental authorities
Forestry authorities are reliable negotiation partners	Distrust in forestry authorities
Entering a conservation agreement is hindered by the fact that the agreement limits the decision-making power of the next owner	Next generation's power of decision retention concerning forest property
Protected sites create a pest risk	Pest risk
My forest property is too small to implement measures for safeguarding biodiversity	Small size of forest property
Practicing active forestry makes it difficult to safeguard biodiversity	Active timber production

External Variables

The explanatory power of attitude, subjective norm, and perceived behavioral control on the intention to safeguard biodiversity among different kinds of forest owners were studied by estimating SEM models separately from the subsamples of the data. The subsamples were based on forest owners' demographics (age, gender, occupation, education, place of residence, residential environment), the size of the forest property, and the implementation of final forest felling during 2010–2014. In addition, separate SEM estimations were calculated for the forest owners who had prior experience of implementing voluntary measures for safeguarding biodiversity in their forests, and those who had not.

Analysis Methods

The frequencies of positive responses to the statements concerning behavioral beliefs and outcome evaluations (Table 1), normative beliefs and motivation to comply (see Variables section) and control beliefs and perceived power to affect (Table 2) were first analyzed to obtain a general idea of how forest owners evaluate these aspects. The descriptive results were computed separately for forest owners with a strong intention and a weak intention to safeguard biodiversity. Pearson's χ^2 test was used to compare the complete percentage distributions among forest owners with a strong intention and a weak intention (Supplement 2). The descriptive analyses were computed with SPSS 29, and the confidence intervals of the percentage shares were calculated with Excel. A two-sided *t*-test was used to compare the means (Field 2013). The explanatory power of attitude, subjective norm, and perceived behavioral control for the intention to safeguard biodiversity was studied using SEM (Bollen 1989). Previous studies of Finnish forest owners were used as a basis for forming statements describing attitude, normative pressures, and perceived constrains (Horne 2006; Horne et al. 2009; Koskela 2011; Mäntymaa et al. 2009). To estimate the model, some variables had to be excluded. The model was formed based on the goodness-of-fit of the model by using the fit indices RMSEA, CFI, and SRMR[1]. For the SEM analysis, following Ajzen (1991), each of the behavioral belief statements was multiplied by the evaluation statement counterparts to generate belief-evaluation products, which could range between 1 and 49. Similarly, each of the pairs of statements concerning the normative beliefs and motivation to comply and the statements concerning control beliefs and the perceived power to affect was multiplied. The products were then analyzed in the SPSS-Amos27 program using SEM (Bollen 1989) (see figure 1). SEM combines factor and multiple regression analysis and is often considered an extension of confirmatory factor analysis, in which the researcher has a theory-based hypothesis (the TPB in this case) for the factor structure in the data (e.g., Bollen 1989; Schreiber et al. 2006). Applying the TPB to empirical data and using SEM allows the analysis of the interaction of attitude toward the specific behavior, subjective norm, and perceived behavioral control, and through them, the intention to perform the behavior. Structural equation models can be used to study how well the theoretical model fits the data (Bollen 1989).

The model is subjected to statistical tests and adequacy tests to investigate its compatibility with empirical observations. In the current study, three commonly used model fit indices are estimated. The critical value of a model's good fit in the study was set as SRMR < 0.08, CFI > 0.9, and RMSEA < 0.08. More conservative values of these measures are also used in the literature: CFI ≥ 0.95 and RMSEA 0.07 (Hooper et al. 2008; Hu and Bentler 1999). SEM was first performed for the entire data and then for subsamples of the data. It was necessary to use subsamples because only a limited number of variables could be included in the simultaneous model estimation. The use of SEM in social sciences has become more widespread, and it has also been applied in studies concerning environmental issues from a landowners' perspective, for example, by Bayard and Jolly (2007); Ficko and Boncina (2015); Karppinen and Berghäll (2015); and Paloniemi and Vainio (2011).

Results

Intention

Half (50%) the respondents indicated that it was likely or very likely (5 to 7 on the scale, strong intention) that they would safeguard biodiversity in their forests during the next 5 years. One-fourth (26%) of the respondents estimated that it was unlikely to very unlikely (3 to 1 on the scale, weak intention) that they would do so. One-fourth (24%) chose option 4 on the scale.

Beliefs and Outcome Evaluations

The descriptive results of the beliefs related to safeguarding biodiversity are presented separately for the respondents with strong intention and weak intention to safeguard biodiversity

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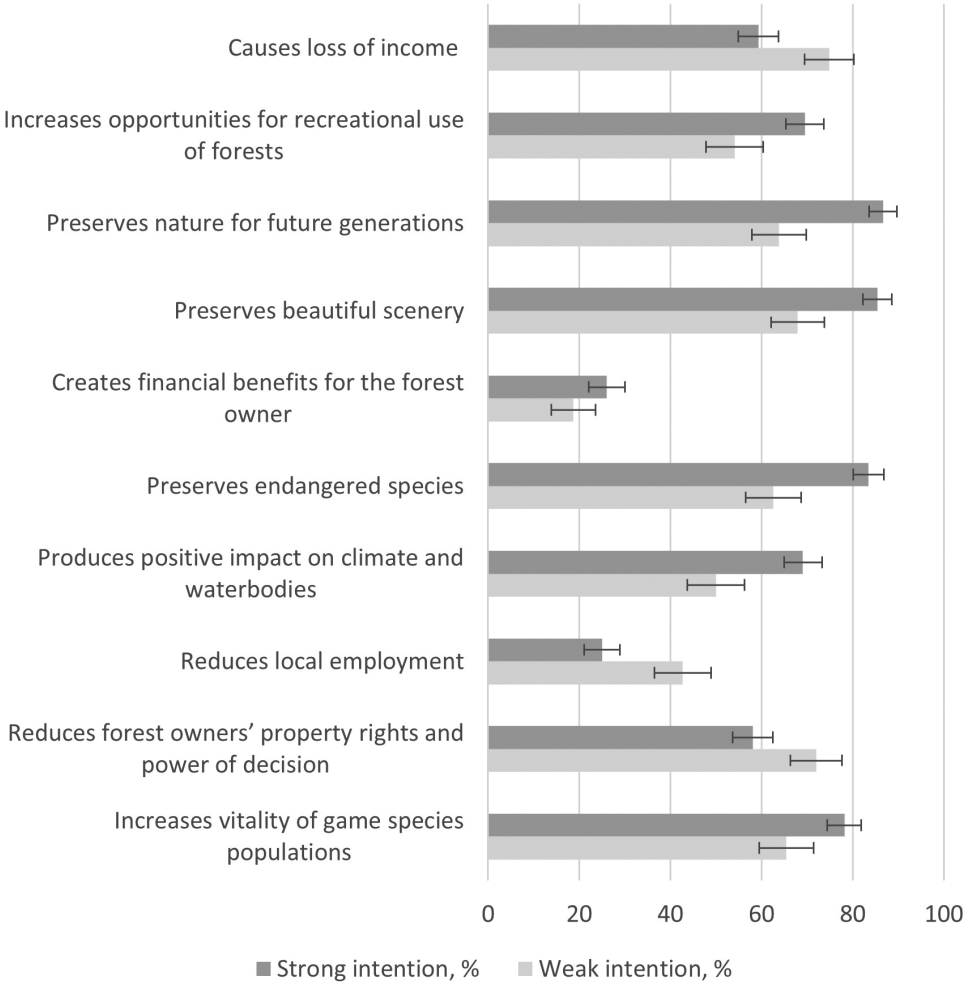


Figure 2 Strength of behavioral beliefs about the effects of safeguarding biodiversity. The percentage of respondents who replied with 5–7 (likely to very likely) among forest owners with strong and weak intentions to safeguard biodiversity in their own forests. The confidence intervals are presented at the 95% confidence level.

in their forests (figure 2). The respondents particularly assessed that safeguarding biodiversity preserves nature for future generations, beautiful scenery, and endangered species (figure 2). In general, the forest owners with a strong intention estimated the positive effects of safeguarding biodiversity to be more likely than the owners with a weak intention. Forest owners with a weak intention assessed the negative effects of safeguarding biodiversity, such as a loss of income or a reduction in ownership rights, as more likely than the forest owners with a strong intention.

The descriptive analysis of the outcome evaluations shows that the importance of retaining property rights and the power of decision and the intergenerational aspect of preserving nature for the next generation were emphasized in both intention groups (figure 3). A larger share of the owners with a strong intention evaluated most of the positive outcomes as important for them more often than the owners with a weak intention. However, the intention groups' assessments were similar regarding the importance of property rights and the effects on local employment.

Normative Beliefs and Motivation to Comply

Especially, the respondents with strong intention believed that representatives of nature conservation organizations and their own family expected them to safeguard biodiversity in their forests (figure 4). They also quite often felt that the advisors of the Finnish Forest Center and the advisors of local Forest Management Associations thought that they should do so.

The respondents in both intention groups were often motivated to comply with the opinions of family members and advisors of local Forest Management Associations. Especially the respondents with a strong intention also indicated motivation to comply with the recommendations of Finnish Forest Center advisors and forestry service entrepreneurs known by the respondent (figure 5). In general, the motivation to comply with the referent groups' expectations was stronger among the owners with a strong intention than those with a weak intention.

Control Beliefs and Perceived Power to Affect

Forest owners in both intention groups often agreed with control belief statements stating that entering a conservation agreement is hindered by the fact that the agreement limits the decision-making power of the next owner, and the forest property was too small to implement measures for safeguarding biodiversity (figure 6). Among the respondents with a strong intention to safeguard forest biodiversity, the

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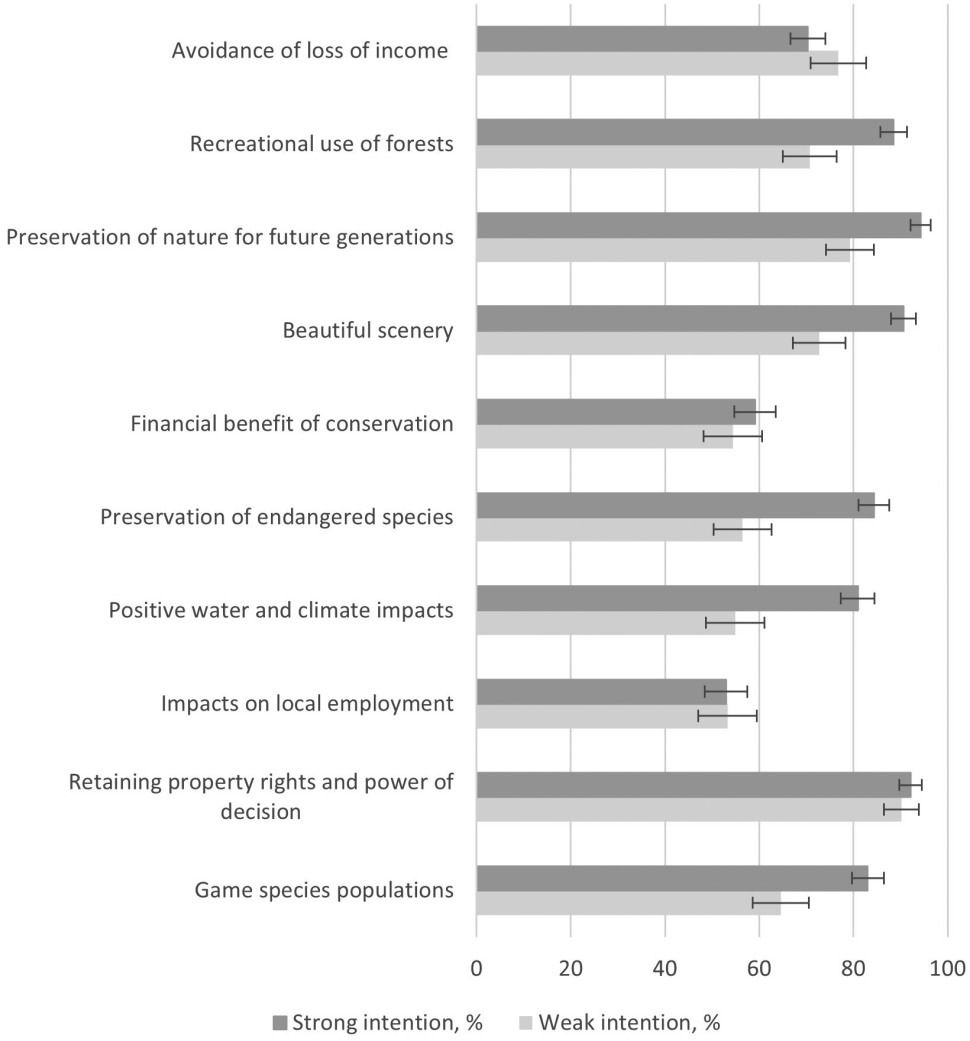


Figure 3 Outcome evaluations of the importance of the factors for the forest owners themselves. The percentage of respondents who replied with 5–7 (important to very important) among forest owners with strong and weak intentions to safeguard biodiversity in their own forests. The confidence intervals are presented at the 95% confidence level.

statements describing facilitating factors were agreed mostly more often than among the forest owners with a weak intention. In contrast, a weak intention was related to a stronger emphasis on negative controlling factors, such as the statement that practicing active forestry made it difficult to safeguard biodiversity (figure 6).

Forest owners' estimations about the impact of a pest risk, the retention of the next owner's power of decision, and the amount of compensation for conservation on their decision-making related to safeguarding biodiversity were rather similar among owners with a strong or weak intention (figure 7).

Factors Explaining Intention to Safeguard Forest Biodiversity: The Structural Equation Model

Whole Sample

The impact of attitudes, subjective norm, and perceived behavioral control on the intention were empirically tested by estimating structural equation models. The SEM estimation was found satisfactory in respect of statistical goodness-of-fit criteria (SRMR = 0.04, CFI = 0.977, RMSEA = 0.043). The estimation results suggest that the impact of controlling factors on the intention to safeguard biodiversity was slightly stronger (-0.31) than the explanatory power of attitude (0.21) and normative pressures (0.20) (figure 8). All four products of control beliefs and perceived power to affect in the model (level of compensation, active timber production, next generation's power of decision retention, small size of forest property) described negative aspects or hindering factors related to the safeguarding of biodiversity. Perceived behavioral control therefore had an inhibiting impact on the intention to safeguard biodiversity, as the negative coefficient shows. The belief-evaluation products in the model depicted the positive effects of safeguarding biodiversity (preserving nature for future generations, preserving beautiful scenery and endangered species, enhancing recreational use of forests, and positive impacts on climate and waterbodies). The referents affecting forest owners' decision-making on the model were the advisors of local Forest Management Associations, the Finnish Forest Center's advisors, forestry service entrepreneurs known by the respondent, and other forest owners known by the respondent.

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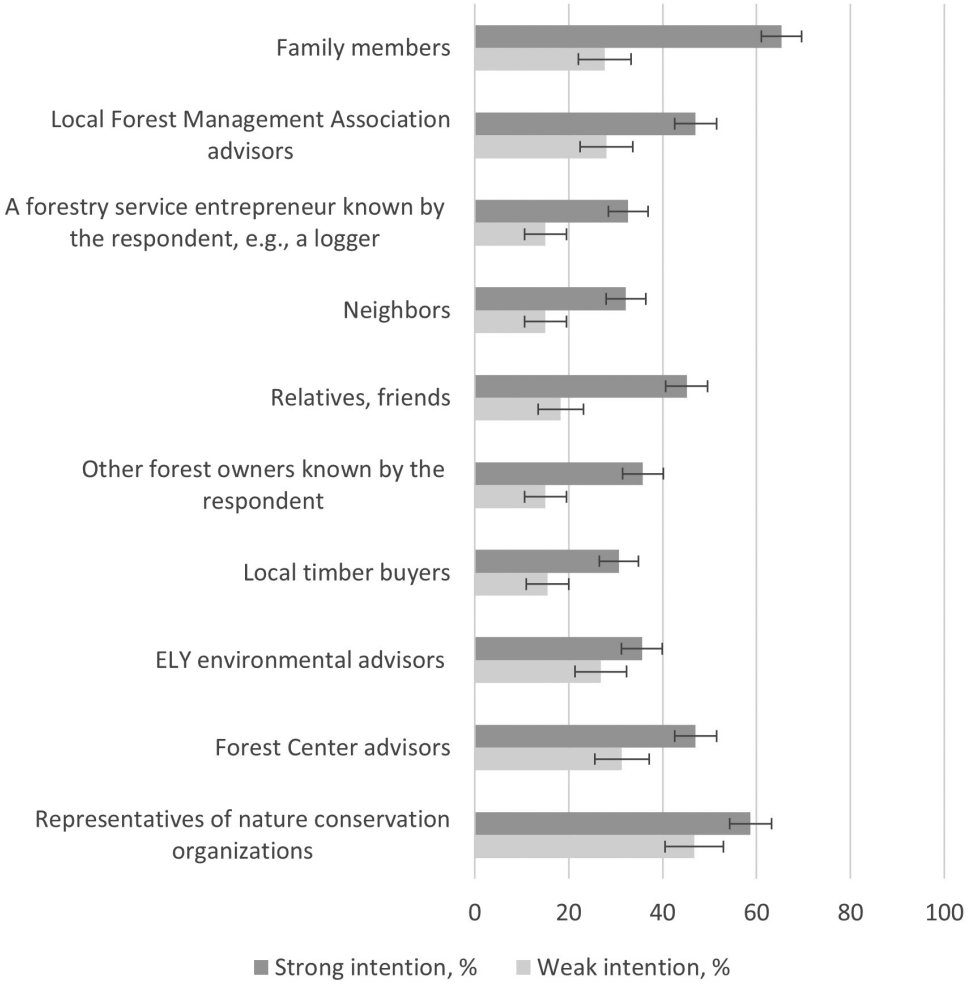


Figure 4 Strength of normative beliefs. The percentage of respondents who replied with 5–7 (the representatives of the referent group think that I should safeguard biodiversity) among forest owners with strong and weak intentions to safeguard biodiversity in their own forests. The confidence intervals are presented at the 95% confidence level.

Subsamples

The structural equation models were estimated for the subsamples of the data (Table 3). The estimation results of subsamples describe the possible differences in explanatory power of attitude, subjective norm, and perceived behavioral control among forest owners with different characteristics.

The estimation results for men and women were somewhat similar regarding the effects of attitude on the intention to safeguard forest biodiversity (Table 3). Controlling factors affected female forest owners slightly more than male forest owners. Men's intention was explained more strongly by normative pressures than women's intention.

Among the youngest age cohort, perceived control and attitude had a similarly strong effect on the intention to safeguard biodiversity. The explanatory power of the subjective norm was weak among forest owners under 45 years of age, but the effect increased with age. In contrast, the role of attitude decreased with age. In the two older age cohorts, the dominant explanatory factor for the intention to safeguard biodiversity was perceived behavioral control.

For those forest owners with an academic degree, the effect of the subjective norm was lower, and the effect of attitude stronger, than among forest owners with a lower educational level. The effect of the subjective norm was especially strong among forest owners without a degree.

For wage earners, perceived control was an especially strong negative factor explaining the intention to safeguard biodiversity. For pensioners, perceived control was also the dominant explanatory factor, but the impact was weaker than among wage earners. The SEM estimations did not perform properly for farmers and forestry entrepreneurs; goodness-of-fit values SRMR and RMSEA were nonsatisfactory. The model estimations further suggest that among absentee forest owners and owners living permanently on the property, the effects of perceived constraints were emphasized. Attitude and normative pressures were both important factors explaining the intention among the forest owners living in population centers. Among city dwellers (>100,000 inhabitants), the impact of attitude appeared to be very low, and the impact of controlling factors to be strong.

Owners of medium-sized forest properties (10 to 49 ha) were particularly affected by perceived controlling factors. For the owners of larger forest holdings (50 to 99 ha), the effect of attitude was emphasized. The estimation results were rather similar among the forest owners who had

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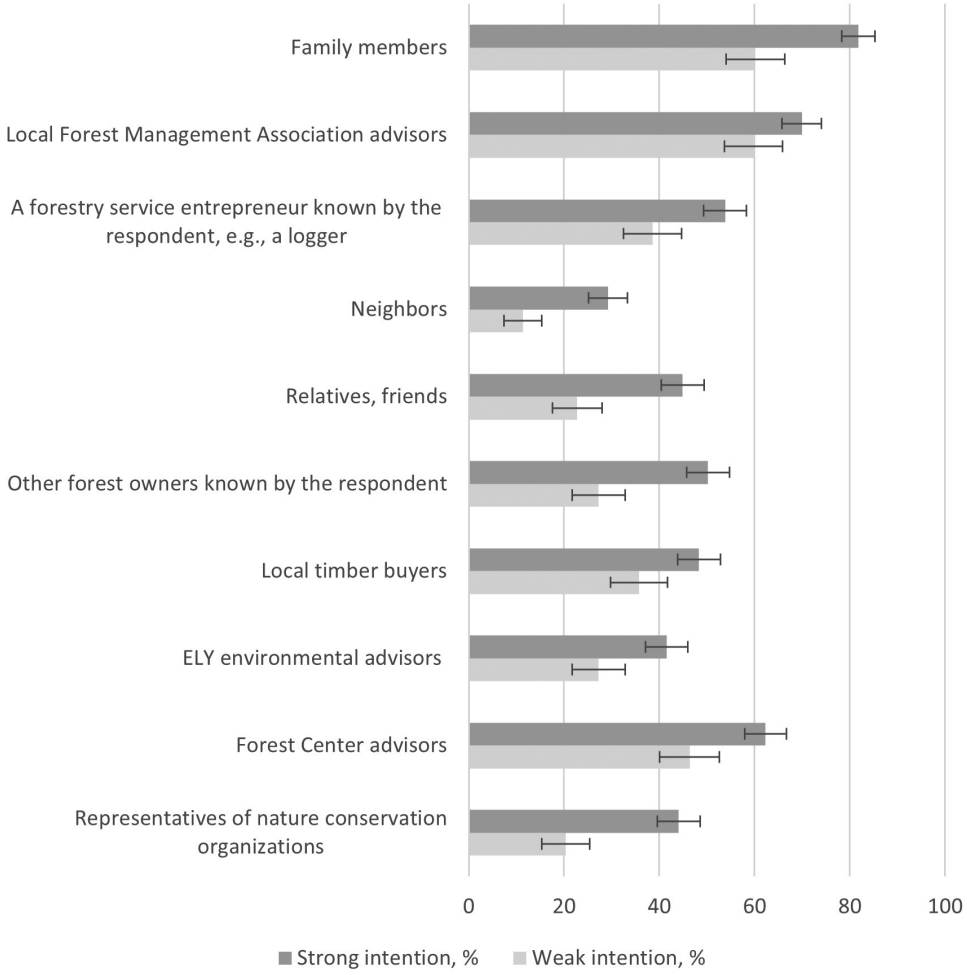


Figure 5 Motivation to comply with the opinions of different referent groups. The percentage of respondents who replied with 5–7 (comply or completely comply) among forest owners with strong and weak intentions to safeguard biodiversity in their own forests. The confidence intervals are presented at the 95% confidence level.

implemented final forest felling and the forest owners who had not. For forest owners with experience of implementation of voluntary measures for safeguarding biodiversity in their forests, the role of normative pressures appeared small.

Discussion

The impact of attitudes, subjective norm, and perceived behavioral control on the intention to safeguard biodiversity were empirically tested by estimating structural equation models. The empirical model estimations supported the TPB.

According to the SEM estimations, the most important decisive factor for intention was perceived behavioral control, which had a slightly stronger explanatory power than attitude or subjective norm. In the model, perceived behavioral control included only inhibiting factors. Therefore, according to forest owners' own assessment, obstacles to the implementation of measures for safeguarding biodiversity are prominent. Forest owners consider the retention of intergenerational decision-making power as an important element in implementing measures for safeguarding biodiversity. This notion is well in line with Raedeke et al. (2001), who found that one of the identified barriers for landowners to participate in voluntary ecosystem management programs was a concern for stronger regulation of ownership. Land-related decisions have been identified as complex and, in some cases, also emotional (Markowski-Lindsay et al. 2016).

In this study, practicing active forestry was considered to hinder the intention to safeguard biodiversity. Many Finnish forest owners emphasize economic goals and consider forest income an essential part of their motivation to own a forest (Karppinen et al. 2020; Takala et al. 2017). Finnish forest owners have also considered timber production to be the most important forest ecosystem service for themselves and for the whole country (Koskela et al. 2021). Safeguarding biodiversity and timber production are not completely mutually exclusive. It is essential that forest owners are aware of the possibilities to apply forest management methods in commercial forests that take biodiversity specifically into account. However, implementing some active nature management practices such as controlled burnings require expertise and certain equipment and often require cost-sharing.

The small size of a forest property was perceived as an obstacle to safeguarding biodiversity. However, implementation of measures for safeguarding biodiversity does not always directly depend on the size of the forest property, and

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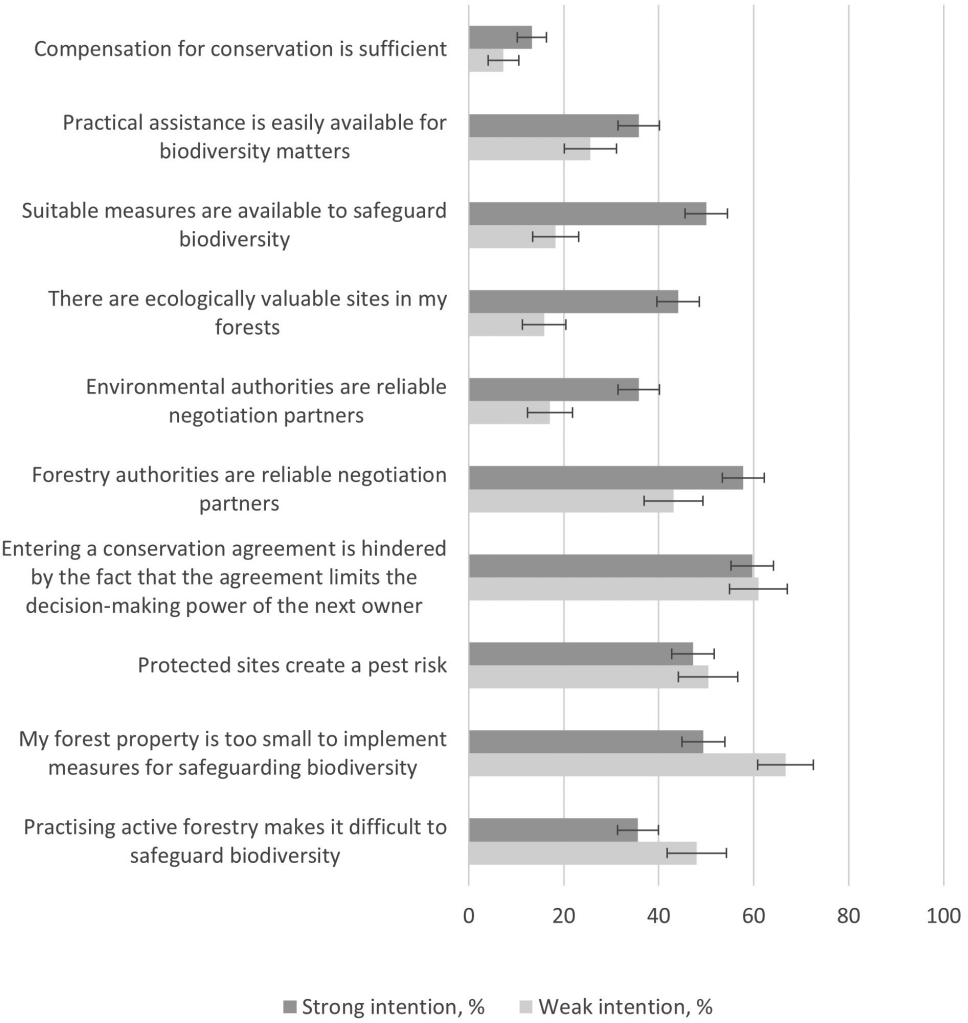


Figure 6 Strength of control beliefs. The percentage of respondents who replied with 5–7 (agree to completely agree) among forest owners with strong and weak intentions to safeguard biodiversity in their own forests. The confidence intervals are presented at the 95% confidence level.

some measures can also be applied on a small forest area. The perceived control factors included the level of compensation in the estimated model. Horne (2006) has identified the importance of contract terms for the level of a compensation claim. The comparison between the value of the compensation and timber production is important for the forest owner.

Safeguarding biodiversity was associated with many positive beliefs, such as preserving nature for future generations, preserving beautiful scenery and endangered species, enhancing recreational use of forests, and positive impacts on climate and waterbodies. Fielding et al. (2005) and Karppinen and Berghäll (2015) identified differences in the strength of beliefs and outcome evaluations between forest owners with strong and weak intention to perform the behavior in question. A similar pattern was found in this study: Forest owners with strong intention agreed more often with the positive effects that safeguard biodiversity creates, and those with weak intention had a stronger belief in the adverse effects. When voluntary conservation mechanisms are developed, the positive associations could be used to support and strengthen the intention of forest owners.

The study results suggest that the roles of forestry professionals and other forest owners are considerable as sources of normative pressures. Local Forest Management Associations and Finnish Forest Center advisors often cooperate closely with forest owners in forest management activities, as do forest entrepreneurs, such as local loggers. Korhonen et al. (2013) have identified the important role Finnish Forest Center representatives play in delivering information about voluntary conservation. It is also important that the whole variety of measures for safeguarding biodiversity are brought to the fore when forest owners are advised in decision-making regarding forests. In the descriptive analysis, the family emerged as a source of normative pressures, especially among forest owners with strong intention. However, this was not confirmed by model estimations.

The demographics of forest owners affected the model results. The effect of the subjective norm increased and the role of attitude decreased with increasing age. The result suggests that older forest owners are more influenced by referents' advice. Karppinen and Berghäll (2015) found partly similar results considering forest owners' intention to implement timber stand improvement. Normative pressures were the dominant explanatory factor in both the oldest age cohort and in the youngest (Karppinen and Berghäll 2015). An academic education seems to increase the importance of attitudes and decrease the role of normative pressures. This

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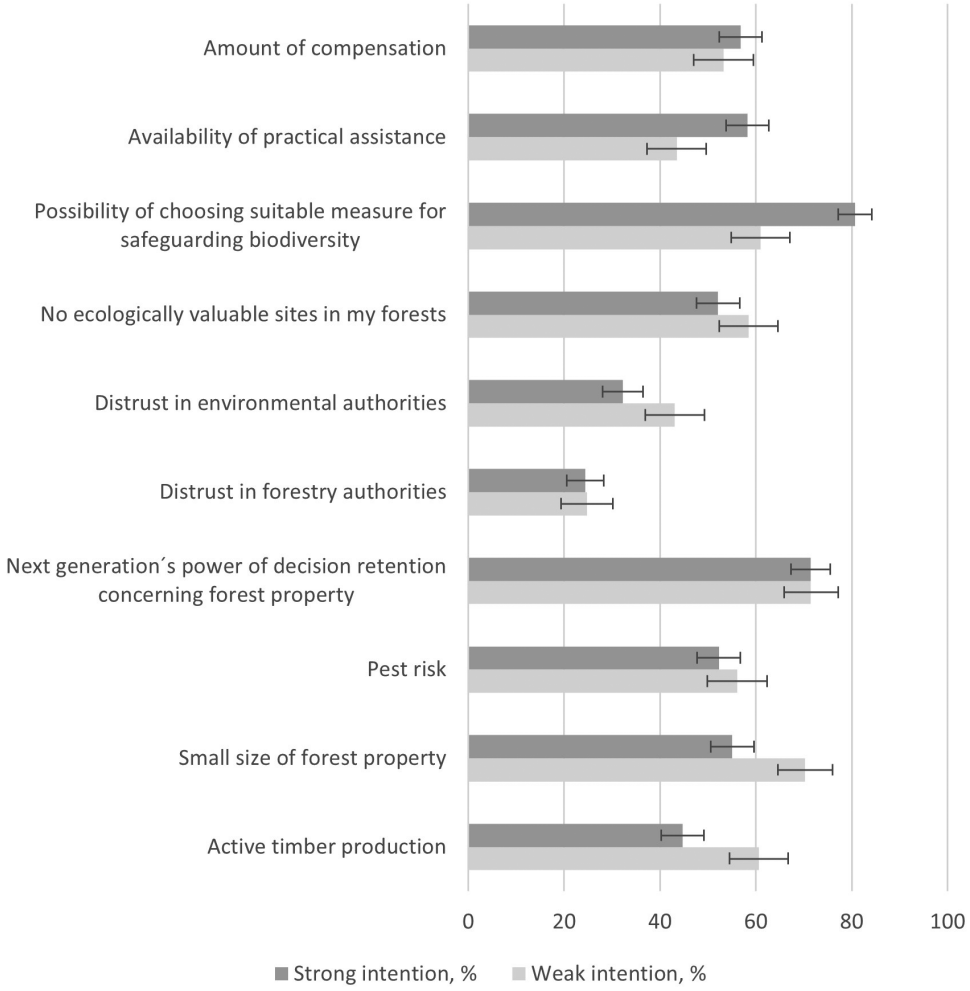


Figure 7 Perceived power to affect of the factors that hinder or facilitate the safeguarding of forest biodiversity. The percentage of respondents who replied with 5–7 (affects or affects very much decision-making) among forest owners with strong and weak intentions to safeguard biodiversity in their own forests. The confidence intervals are presented at the 95% confidence level.

finding compares well with Karppinen and Berghäll (2015). Younger age and higher level of education have been found to be associated with stronger interest in implementing measures that increase biodiversity (Husa and Kosenius 2021).

The role of attitude appeared to be greater for owners of larger forest holdings (50–99 ha). A large forest holding probably offers more opportunities to carry out various forest management practices and also measures for safeguarding biodiversity. Furthermore, it is more likely that a large area includes ecologically valuable sites while allowing the possibility to practice active forestry outside these areas. Similarly, Danley (2018) suggests that a large forest property is positively associated with leaving voluntary set-asides, which are managed in such way that the natural, cultural, or social value of the stand is not damaged. The Finnish National Forest Strategy 2025 (Ministry of Agriculture and Forestry 2019) includes a goal of increasing the average size of forest properties and thereby enhancing active and business-like forestry. On the other hand, based on the result that a small forest property was perceived as an obstacle to the implementation of conservation measures, increasing forest property sizes could also contribute to safeguarding biodiversity.

The comparison of the background characteristics of the study respondents with those of the respondents of the recent Forest Owner 2020 survey (Karppinen et al. 2020) revealed no large differences. It can therefore be concluded that the data used in this study are relevant to describing Finnish forest owners and their views, although they were collected at the end of 2015. The forest ownership structure is changing in Finland, but the change has been relatively slow. For example, the proportion of farmer forest owners living on their forest property has been decreasing and the proportion of urban forest owners increasing, but the latter trend has been weaker than expected in the last 10 years. The share of well-educated forest owners has also increased between 2009 and 2019 (Hänninen et al. 2011; Karppinen et al. 2020).

Some limitations of the study must be acknowledged. The analysis of nonresponse revealed that a larger proportion of the respondents had a positive attitude toward increasing the level of safeguarding biodiversity in private forests in general than those who did not reply. Although it has been suggested in the study literature (e.g., Sheeran 2002) that intention predicts behavior, further research is needed on the relationship between intention and the actual implementation of the measures for safeguarding biodiversity. Typically, the analysis in the current study was restricted by a lack of knowledge of forest owners' actual behavior. Will they implement the measures for safeguarding forest biodiversity as they intended?

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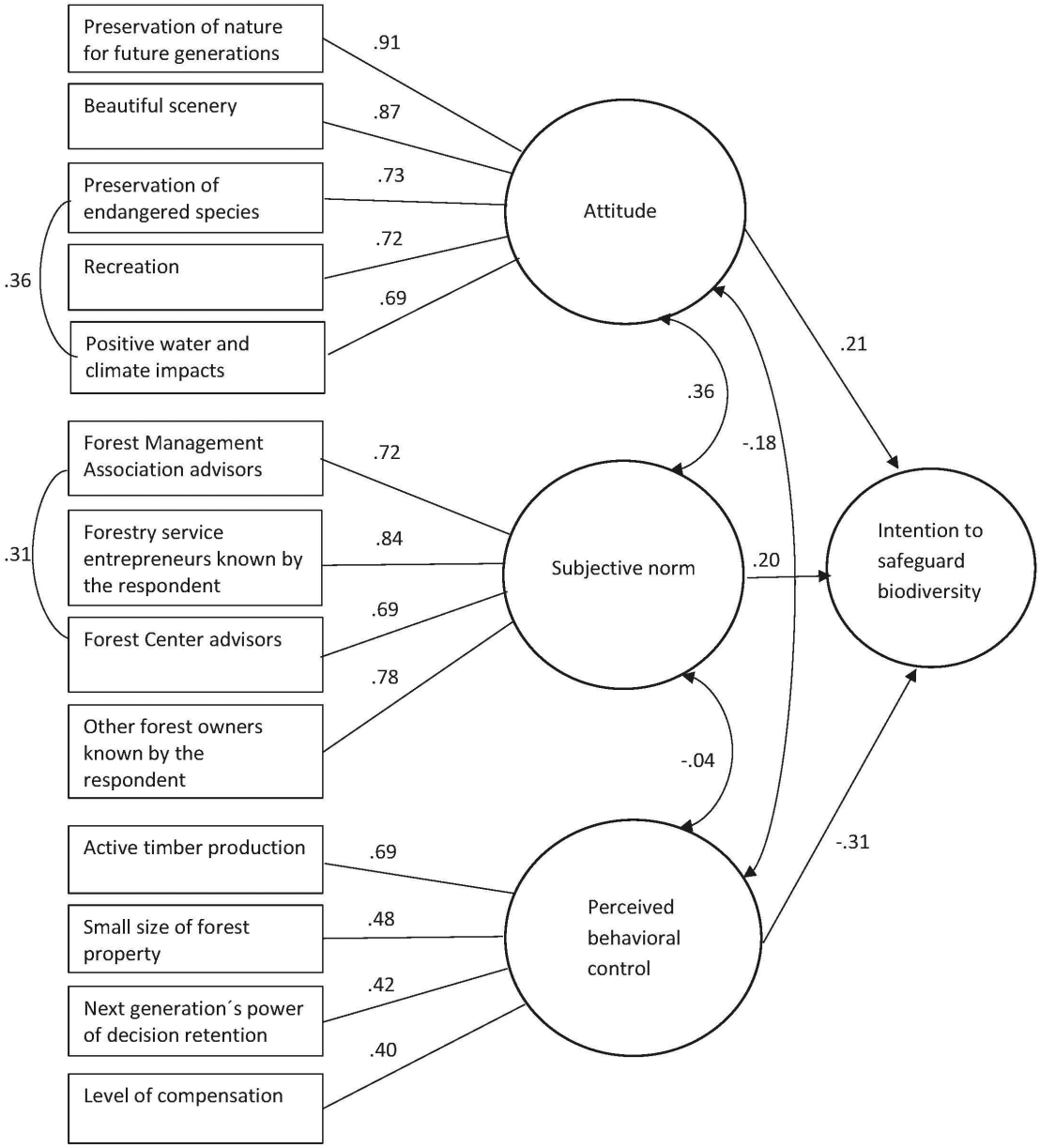


Figure 8 SEM estimation results of the TPB model explaining forest owners' intention to safeguard biodiversity in their own forests (n = 953).

The results highlight the importance of the availability of several kinds of measures for safeguarding biodiversity, including those that allow forest owners to preserve the decision-making power of the future owners of the property. It would be beneficial to further develop methods that combine the enhancement of forest biodiversity and the production of forest ecosystem services, such as carbon sequestration and storage and timber. The use of various map and inventory data to identify ecologically valuable sites can support conservation planning (e.g., Mikkonen et al. 2023) and increase forest owners' awareness about the ecologically valuable sites in their forests. Forest owners should be provided with accurate information about the measures available and applicable on their forest property, and about compensation and contract terms. Karppinen et al.'s (2018) study of forest owners' views on storing carbon in their forests found that if the willingness to take measures was not strongly value-driven, forest owners' participation in the program was vulnerable to the conditions of the process.

Conclusions

The loss of forest biodiversity has generally been recognized in Finland (e.g., Hyvärinen et al. 2019). The EU Biodiversity Strategy for 2030 sets new goals for biodiversity conservation (European Commission 2021; Wolfslehner et al. 2020). That emphasizes the need to lower the threshold for forest owners to implement measures for safeguarding biodiversity. When developing voluntary measures for safeguarding biodiversity in private forests, special attention should be paid to the reduction of the obstacles for the implementation perceived by forest owners.

An environmental forestry subsidy agreement or an establishment of a permanent nature reserve are not decisions that

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Table 3. SEM estimations explaining the intention to safeguard forest biodiversity in various subsamples (nonsatisfactory goodness-of-fit values underlined).

	Coefficient			Goodness-of-fit			
	Attitude	Subjective norm	Perceived control	SRMR	CFI	RMSEA	*N*
Whole data	**0.21**	**0.20**	**-0.31**	**0.04**	**0.977**	**0.043**	**953**
Gender							
Female	0.23	0.11	-0.36	0.061	0.967	0.052	343
Male	0.24	0.24	-0.29	0.042	0.975	0.045	580
Age, years							
-44	0.30	0.09	-0.30	0.069	0.978	0.042	110
45–64	0.27	0.18	-0.34	0.045	0.976	0.045	401
65–	0.13	0.26	-0.32	0.052	0.971	0.046	399
Education							
No degree	0.14	0.34	-0.33	0.061	0.986	0.032	169
Vocational education	0.23	0.19	-0.25	0.046	0.970	0.048	368
Higher vocational diploma	0.07	0.24	-0.39	0.046	0.972	0.048	243
Academic degree	0.28	0.10	-0.36	0.077	0.931	0.077	140
Occupation							
Wage-earner	0.18	0.24	-0.50	0.050	0.966	0.050	305
Farmer or forestry entrepreneur	0.23	0.22	-0.21	0.092	0.932	0.090	89
Entrepreneur	0.43	-0.01	-0.08	0.078	0.962	0.064	55
Pensioner	0.19	0.24	-0.29	0.049	0.977	0.042	439
Place of residence							
Permanently on the property	0.23	0.22	-0.38	0.044	0.979	0.042	361
Elsewhere in same municipality as the property	0.19	0.20	-0.14	0.055	0.948	0.065	189
Outside the property's municipality	0.19	0.22	-0.40	0.051	0.975	0.044	359
Residential environment							
Rural	0.18	0.17	-0.39	0.048	0.972	0.047	475
Population center	0.35	0.29	0.04	0.048	0.979	0.041	160
Town	0.18	0.22	-0.44	0.066	0.971	0.050	162
City > 100,000	0.08	0.22	-0.52	0.066	0.970	0.049	134
Size of forest property, ha							
<10	0.24	0.30	-0.28	0.067	0.969	0.051	133
10–19	0.15	0.20	-0.42	0.048	0.986	0.034	203
20–49	0.19	0.23	-0.40	0.060	0.960	0.058	286
50–99	0.38	0.09	-0.25	0.059	0.954	0.056	166
100–	0.25	0.21	-0.09	0.074	0.950	0.069	94
Final felling during last 5 years							
Yes	0.17	0.20	-0.32	0.045	0.978	0.038	381
No	0.20	0.21	-0.31	0.041	0.984	0.038	390
Experience of voluntarily implementing measures for safeguarding biodiversity*							
No	0.28	0.18	-0.24	0.040	0.974	0.049	549
Yes	0.18	0.09	-0.26	0.051	0.984	0.033	261

*Voluntary implementation of at least one measure to safeguard biodiversity in one's own forest, such as voluntary fixed-term conservation contract, voluntary establishment of private nature reserve, selling of forest area for conservation purposes (voluntarily), exclusion of a forest site or sites from timber harvesting by owners' own decision, participation to a nature management project, taking biodiversity specifically into account in forest management practices. The critical value of a model's good fit in the study was set as SRMR < 0.08, CFI > 0.9, and RMSEA < 0.08. RMSEA, root mean square error of approximation; SRMR, standardized root mean squared residual; CFI, comparative fit index.

forest owners very frequently make. The habituation of such a decision is therefore quite unlikely. Due to the ecological criteria of the sites protected in the METSO program and the budget limitations to implementing these measures, an owner does not always necessarily have the option of establishing a private nature reserve area with compensation or by signing a subsidy agreement. However, forest owners more frequently make decisions about forest management, and they may have different options to take biodiversity into account in these decisions.

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Conflict of Interest

None declared.

Endnotes

[1]The root mean square error of approximation (RMSEA), the standardized root mean squared residual (SRMR) and comparative fit index (CFI) (e.g., Hu and Bentler 1999; McDonald and Ho 2002; Hooper et al. 2008).

Supplementary Materials

Supplementary data are available at *Forest Science* online.

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